# SECTION 1

## SETS AND BINOMIAL EXPANSIONS

# Modelling with Algebra

## Application of Algebra

### Introduction

This section is a continuation of what was discussed in Year 1, sets and binomial expansion. De Morgan's Laws is an important theorem in sets as its application transcends beyond union, intersection and complements. These laws are central to Boolean Algebra where it is used to simplify expressions. In Digital Circuit Design, it is used to transform logic gates with the intention of not only optimising digital circuitry but also reducing cost and saving space. In Computer programming, it is used to simplify logical expressions in code, which can make code cleaner, simpler, and more readable. Database systems use these laws to optimise queries in SQL (Structured Query Language), which enhances data retrieval more efficiently.

The Binomial Theorem is a strong tool that makes the process of expanding binomial expressions much simpler. In developing this theorem, you further your algebra skills and prepare yourselves for higher classes of mathematics. The Binomial Theorem is applied in finance, investment, probability, statistics, computer science, genetics, engineering, game theory, marketing, and physics. It helps in calculating compound interest, risk assessment, modelling success/failure outcomes, understanding data structure complexities, predicting inheritance probabilities and enhancing decision-making in real-world scenarios.

**KEY IDEAS**

- **Binomial Coefficients:** The coefficients ${}^nC_r$ represent the number of ways to choose $r$ elements from $n$ elements and can be found using Pascal's Triangle.

$$ {}^nC_r = \frac{n!}{(n-r)!\,r!} $$

- **Terms in the Expansion:** Each term in the expansion is of the form:

$$ {}^nC_r\, a^{n-r} b^r $$

- The Binomial Theorem provides a formula for expanding expressions of the form $(a+b)^n$ where $n$ is a non-negative integer.
- The complement of an empty set ($\phi$) is equal to the universal set ($\mu$), i.e. $\phi' = \mu$.
- The intersection of a set $A$ and its complement ($A'$) is equal to the null set ($\varnothing$), i.e. $A \cap A' = \varnothing$.
- The union of a set $A$ and its complement ($A'$) is equal to the universal set ($\mu$), i.e. $A \cup A' = \mu$.

# 1 Establishing De Morgan's Laws of Set Theory

De Morgan's Laws are rules which are used to relate intersections and unions through complements. Generally, according to De Morgan's Laws:

1. The complement of the union of sets is equal to the intersection of the individual complements. This means that if $A$, $B$ and $C$ are sets,

$$ (A \cup B)' = A' \cap B', \qquad (A \cup B \cup C)' = A' \cap B' \cap C'. $$

In words: *the complement of the union equals the intersection of the individual complements.*

2. The complement of the intersection of sets is equal to the union of the individual complements. This means that if $A$, $B$ and $C$ are sets,

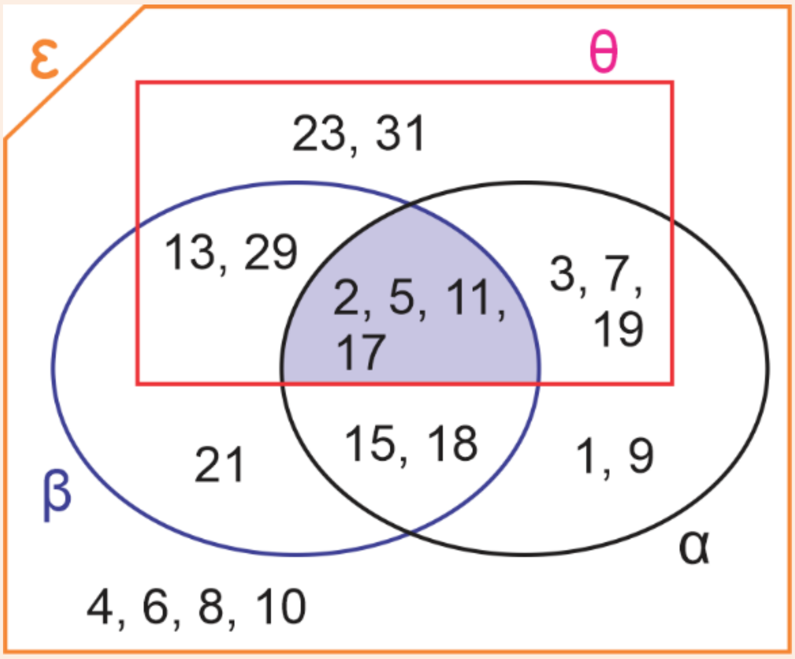
$$(A \cap B)' = A' \cup B', \qquad (A \cap B \cap C)' = A' \cup B' \cup C'.$$

In words: *the complement of the intersection equals the union of the individual complements.*

Let us use the activity below to validate these rules.

**Activity 1.1: De Morgan's Laws**

1. Make a copy of the diagram below:




2. List the elements in $\alpha$, $\beta$ and $\theta$.
3. List the elements in $\alpha'$, $\beta'$ and $\theta'$.
4. Find (a) $(\alpha \cup \beta \cup \theta)'$ (b) $\alpha' \cap \beta' \cap \theta'$ (c) $(\alpha \cap \beta \cap \theta)'$ (d) $\alpha' \cup \beta' \cup \theta'$.
5. Compare your result in 4(a) to 4(b). What conclusion can you draw?
6. Compare your result in 4(c) to 4(d). What conclusion can you draw?

Compare your answer to the suggested solution below.

2.

$$\begin{aligned}\alpha &= \{1, 2, 3, 5, 7, 9, 11, 15, 17, 18, 19\},\\ \beta &= \{2, 5, 11, 13, 15, 17, 18, 21, 29\},\\ \theta &= \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31\}.\end{aligned}$$

3.

$$\alpha' = \{4, 6, 8, 10, 13, 21, 23, 29, 31\},$$
$$\beta' = \{1, 3, 4, 6, 7, 8, 9, 10, 19, 23, 31\},$$
$$\theta' = \{1, 4, 6, 8, 9, 10, 15, 18, 21\}.$$

4.

(a) $(\alpha \cup \beta \cup \theta)' = \{4, 6, 8, 10\}$

(b) $\alpha' \cap \beta' \cap \theta' = \{4, 6, 8, 10\}$

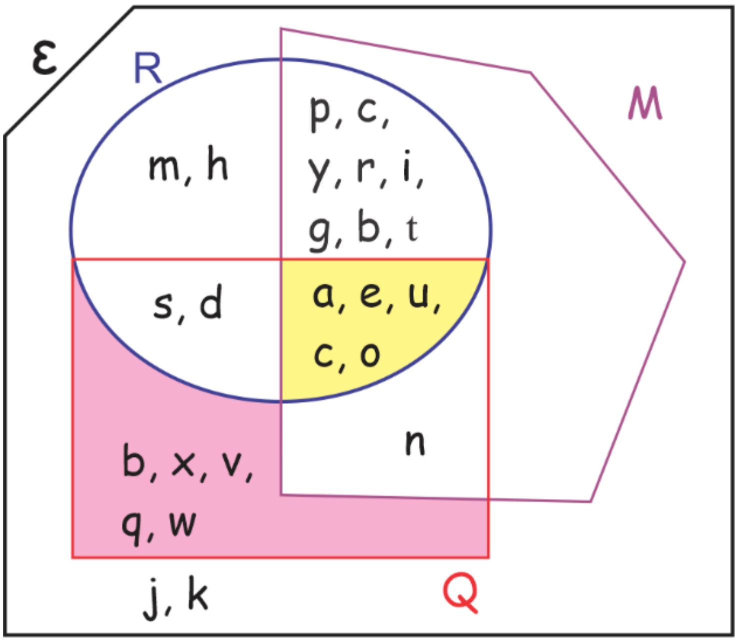
(c) $(\alpha \cap \beta \cap \theta)' = \{1, 3, 4, 6, 8, 9, 10, 13, 19, 21, 23, 29, 31\}$

(d) $\alpha' \cup \beta' \cup \theta' = \{1, 3, 4, 6, 8, 9, 10, 13, 19, 21, 23, 29, 31\}$

5. Results in 4(a) and 4(b) are equal. This confirms De Morgan's Law.
6. Results in 4(c) and 4(d) are equal. This confirms De Morgan's Law.

**Example 1.1**

Study the diagram carefully and use it to answer the questions.




1. List the elements in $R$, $M$, $Q$ and $\varepsilon$.
2. Show that $R' \cap M' = (R \cup M)'$.
3. Show that $(R \cap M \cap Q)' = R' \cup M' \cup Q'$.

**Solution**

1.

$$R = \{s, u, b, d, e, r, m, a, t, o, g, l, y, p, h, i, c\},$$
$$M = \{u, n, c, o, p, y, r, i, g, h, t, a, b, l, e\} \text{ and}$$
$$Q = \{a, e, u, c, o, n, x, q, v, w\} \text{ are subsets of}$$
$$\varepsilon = \{s, u, b, d, e, r, m, a, t, o, g, l, y, p, h, i, c, n, x, q, v, w, j, k\}.$$

2. $R' \cap M' = (R \cup M)'$

   Let us first solve for $R' \cap M'$:

$$\{b, x, v, n, q, w, j, k\} \cap \{m, h, s, d, b, x, v, q, w, j, k\}$$

$$R' \cap M' = \{b, x, v, q, w, j, k\} \quad \ldots\ldots (1)$$

   Next, let us solve for $(R \cup M)'$:

$$(R \cup M)' = \{s, u, b, d, e, r, m, a, t, o, g, l, y, p, h, i, c, n\}'$$

$$(R \cup M)' = \{b, x, v, q, w, j, k\} \quad \ldots\ldots (2)$$

   Since equation (1) is equal to equation (2), it shows that $R' \cap M' = (R \cup M)'$.

   *(Note: the original textbook's working reproduced above actually contains an arithmetic slip — from the given $R$, $M$ and $\varepsilon$, the correct complements are $R' = \{j, k, n, q, v, w, x\}$ and $M' = \{d, j, k, m, q, s, v, w, x\}$* (no 'b' or 'h'), *giving $R' \cap M' = (R \cup M)' = \{j, k, q, v, w, x\}$ rather than the $\{b, x, v, q, w, j, k\}$ shown above. The error appears on both sides of the equation, so it does not affect the conclusion that $R' \cap M' = (R \cup M)'$, but the specific set of elements shown is incorrect.)*

3. Let's first solve for $(R \cap M \cap Q)'$:

$$(R \cap M \cap Q)' = (a, e, u, c, o)'$$

$$= \{s, b, d, r, m, t, g, l, y, p, h, i, n, x, q, v, w, j, k\} \quad \ldots\ldots (1)$$

   Next, let's find $R' \cup M' \cup Q'$:

$$\begin{aligned} R' \cup M' \cup Q' &= \{b, x, v, n, q, w, j, k\} \cup \{m, h, s, d, b, x, v, q, w, j, k\} \\ &\quad \cup \{m, h, p, c, y, r, i, g, b, t, j, k\} \\ &= \{s, b, d, r, m, t, g, l, y, p, h, i, n, x, q, v, w, j, k\} \quad \ldots\ldots (2) \end{aligned}$$

   Since equation (1) is equal to equation (2), it shows $(R \cap M \cap Q)' = R' \cup M' \cup Q'$.

   *(Note: this working carries the same erroneous $R'$ and $M'$ from part 2, plus a similarly erroneous $Q' = \{b, c, g, h, i, j, k, m, p, r, t, y\}$ instead of the correct $Q' = \{b, d, g, h, i, j, k, l, m, p, r, s, t, y\}$* (no 'c'; includes 'd', 'l', 's'). *Taking the union of the textbook's own three (erroneous) complements gives $\{b, c, d, g, h, i, j, k, m, n, p, q, r, s, t, v, w, x, y\}$ — note the 'c' — not the $\{s, b, d, r, m, t, g, l, y, p, h, i, n, x, q, v, w, j, k\}$ (with an 'l' instead of a 'c') printed above, so the final line does not even follow from the textbook's own stated $R'$, $M'$, $Q'$. The correct result, computed directly from $R \cap M \cap Q = \{a, c, e, o, u\}$, is $(R \cap M \cap Q)' = \varepsilon \setminus \{a, c, e, o, u\} = \{b, d, g, h, i, j, k, l, m, n, p, q, r, s, t, v, w, x, y\}$, which does agree with $R' \cup M' \cup Q'$ once the correct complements from the note above are used instead. As before, the identity itself still holds; only the specific elements printed in the textbook's arithmetic are unreliable.)*

## 2 Applying De Morgan's Laws of Set Theory

We have learned about De Morgan's Laws of Set Theory. Now, we will look at its application. Before we consider this application, let's familiarise ourselves with a few properties of complements as it will be central to our understanding.

### Properties of Complement

1. **Union of a set and its complement.** The union of a set $A$ and its complement ($A'$) is equal to the universal set ($\mu$), i.e. $A \cup A' = \mu$.
2. **Intersection of a set and its complement.** The intersection of a set $A$ and its complement ($A'$) is equal to the null set ($\varnothing$), i.e. $A \cap A' = \{\ \} = \phi$.
3. **Double complement.** The complement of the complement of set $A$ will give the same set, $A$, i.e. $(A')' = A$.
4. **Complement of an empty set.** The complement of an empty set ($\phi$) is equal to the universal set ($\mu$), i.e. $\phi' = \mu$.
5. **Complement of a universal set.** The complement of the universal set is equal to the null set, i.e. $\mu' = \phi$.

**Example 1.2**

Rewrite: a. $(X' \cup Y)'$ b. $(X \cap Y')'$

**Solution**

a. $(X' \cup Y)'$

Recall that from De Morgan's laws, we have $(A \cup B)' = A' \cap B'$. So, for $(X' \cup Y)'$, we will replace $A$ with $X'$ and $B$ with $Y$. This implies $(X' \cup Y)' = (X')' \cap Y'$. From the double complement property, $(X')' = X$. Therefore, $(X' \cup Y)' = X \cap Y'$.

b. $(X \cap Y')'$

Recall that from De Morgan's laws, we have $(A \cap B)' = A' \cup B'$. So, for $(X \cap Y')'$, we will replace $A$ with $X$ and $B$ with $Y'$. This implies $(X \cap Y')' = X' \cup (Y')'$. From the double complement property, $(Y')' = Y$. Therefore, $(X \cap Y')' = X' \cup Y$.

**Example 1.3**

Show that:

a. $(A \cup B' \cup C')' = A' \cap B \cap C$

b. $(A' \cap B \cap C')' = A \cup B' \cup C$

**Solution**

a. Recall that $(A \cup B \cup C)' = A' \cap B' \cap C'$. This means, $(A \cup B' \cup C')' = A' \cap (B')' \cap (C')' = A' \cap B \cap C$.

b. Recall that $(A \cap B \cap C)' = A' \cup B' \cup C'$. So, $(A' \cap B \cap C')' = (A')' \cup B' \cup (C')' = A \cup B' \cup C$.

**Example 1.4**

Rewrite $\left[(P \cup Q')' \cap P\right]'$.

**Solution**

**Method 1** — Let us start with the outer bracket.

Using $[A \cap B]' = A' \cup B'$:

$$\left[(P \cup Q')' \cap P\right]' = (P \cup Q')'' \cup P' = (P \cup Q') \cup P'$$

Since the union is commutative, $P \cup Q' = Q' \cup P$:

$$\left[(P \cup Q')' \cap P\right]' = (Q' \cup P) \cup P'$$

Since the union of sets is associative, $(Q' \cup P) \cup P' = Q' \cup (P \cup P')$:

$$\left[(P \cup Q')' \cap P\right]' = Q' \cup (P \cup P')$$

Recall that $P \cup P' = \mu$ (universal set):

$$\left[(P \cup Q')' \cap P\right]' = Q' \cup \mu = \mu$$

**Method 2** — We will start with the inner bracket and work our way out.

$$\left[(P \cup Q')' \cap P\right]'$$

$$(P \cup Q')' = P' \cap Q'' = P' \cap Q$$

$$\left[(P \cup Q')' \cap P\right]' = \left[(P' \cap Q) \cap P\right]'$$

Since the intersection of sets is commutative, $P' \cap Q = Q \cap P'$:

$$\left[(P \cup Q')' \cap P\right]' = \left[(Q \cap P') \cap P\right]'$$

Since intersection is associative, $(Q \cap P') \cap P = Q \cap (P' \cap P)$:

$$\left[(P \cup Q')' \cap P\right]' = \left[Q \cap (P' \cap P)\right]'$$

Recall that $P' \cap P = \varnothing$:

$$\left[(P \cup Q')' \cap P\right]' = [Q \cap \varnothing]' = [\varnothing]' = \mu$$

**Example 1.5**

Using the Venn diagram, shade the region represented by:

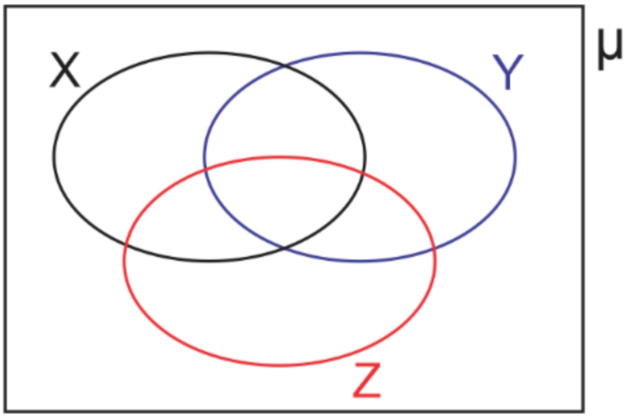
a. $(X \cup Y)' \cap Z$

b. $X' \cap (Y \cup Z)$

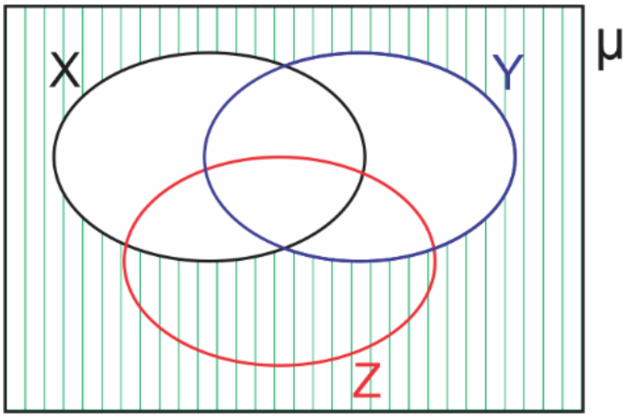
**Solution**

**a.** $(X \cup Y)' \cap Z$

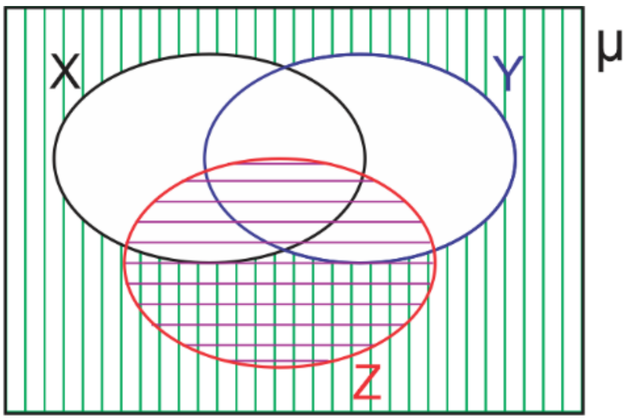
Since the expression involves three sets, $X$, $Y$ and $Z$, we will construct a three-set Venn diagram. We will represent the universal set with $\mu$. An example is shown below:



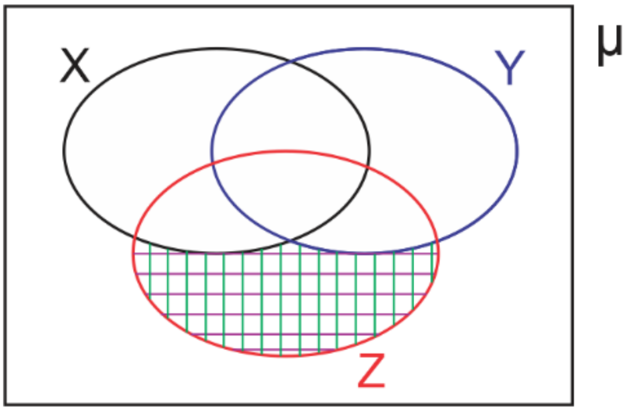

$(X \cup Y)'$ represents elements which are not in $X \cup Y$. We use vertical lines to shade this region:




Then, on the same diagram, set $Z$ is shaded with horizontal lines. The intersection of $(X \cup Y)'$ and $Z$ is the double-shaded (cross-hatched) area:




Therefore, $(X \cup Y)' \cap Z$ is the region shown below, shaded in a single cross-hatch pattern:




**b.** $X' \cap (Y \cup Z)$ is found in the same way: shade $X'$ (everything outside circle $X$) with vertical lines, shade $Y \cup Z$ with horizontal lines, and the region shaded by both patterns is the required set.

In year 1, we learned about set theory laws. Examples of these laws include commutative, associative and distributive. We also learned to identify the regions in a three-set Venn diagram. In this lesson, we will apply these concepts to answer real-life problems.

## How to complete a three-set Venn diagram

1. Start with the intersection of the three sets and proceed to the regions involving two sets.
2. Then work on the regions involving one set and finally the regions in the universal set that do not intersect with any of the three sets.

To summarise, start from the centre (the region with the most overlaps) and work your way out.

**Example 1.6**

Set $A = \{\text{odd numbers less than } 15\}$
Set $B = \{W : 4 < W \leq 11\}$
Set $D = \{\text{Prime numbers less than } 14\}$
All of which are subsets of Set $R = \{0, 1, 2, 3, 4, \ldots, 16\}$.

a. List the members of the following sets: (i) $A$ (ii) $B$ (iii) $D$ (iv) $A \cup B$ (v) $(A \cup B) \cap D'$ (vi) $(A' \cap B) \cup D$.

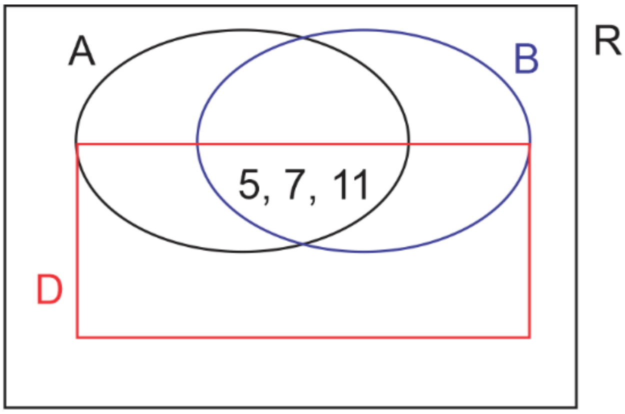
b. Represent $A$, $B$, $D$ and $R$ using a Venn diagram.

**Solution**

a.

$$
\begin{aligned}
\text{(i)} \quad A &= \{1,3,5,7,9,11,13\} \\
\text{(ii)} \quad B &= \{5,6,7,8,9,10,11\} \\
\text{(iii)} \quad D &= \{2,3,5,7,11,13\} \\
\text{(iv)} \quad A \cup B &= \{1,3,5,7,9,11,13\} \cup \{5,6,7,8,9,10,11\} \\
&= \{1,3,5,6,7,8,9,10,11,13\} \\
\text{(v)} \quad (A \cup B) \cap D' &= \{1,3,5,6,7,8,9,10,11,13\} \cap \{1,4,6,8,9,10,12,14,15,16\} \\
&= \{1,6,8,9,10\} \\
\text{(vi)} \quad A' \cap B &= \{2,4,6,8,10,12,14,15,16\} \cap \{5,6,7,8,9,10,11\} = \{6,8,10\} \\
(A' \cap B) \cup D &= \{6,8,10\} \cup \{2,3,5,7,11,13\} = \{2,3,5,6,7,8,10,11,13\}
\end{aligned}
$$

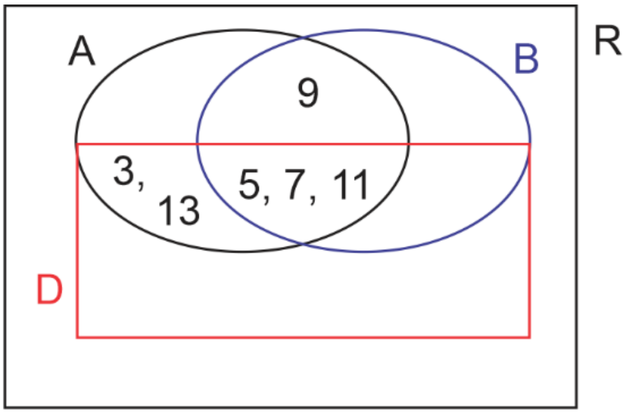
b. We first find the triple intersection: $A \cap B \cap D = \{5,7,11\}$.




Then proceed to the pairwise intersections. There are three of these:

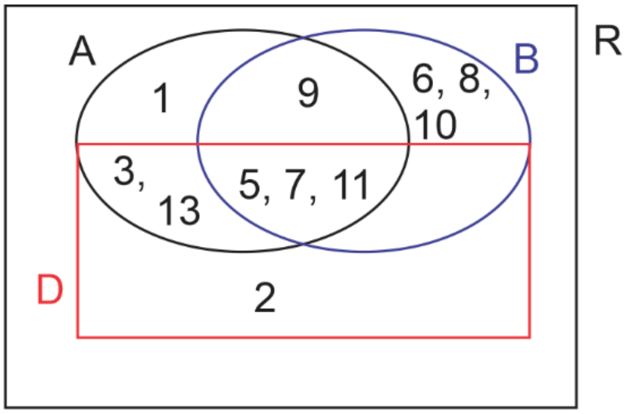
$$
\begin{aligned}
A \cap B \text{ only} &= A \cap B \cap D' = \{9\} \\
A \cap D \text{ only} &= A \cap B' \cap D = \{3,13\} \\
B \cap D \text{ only} &= A' \cap B \cap D = \{\ \}
\end{aligned}
$$

Let's update our diagram with these values:



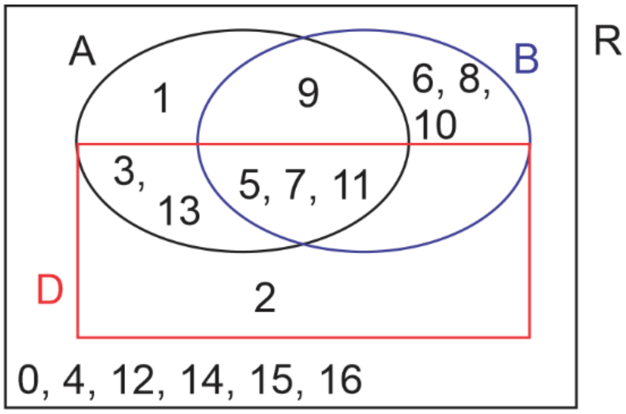

Next, find the single intersections. There are three of them:

$$
\begin{aligned}
A \text{ only} &= \{1\} \\
B \text{ only} &= \{6,8,10\} \\
D \text{ only} &= \{2\}
\end{aligned}
$$




Finally, we find $(A \cup B \cup D)' = \{0, 4, 12, 14, 15, 16\}$.

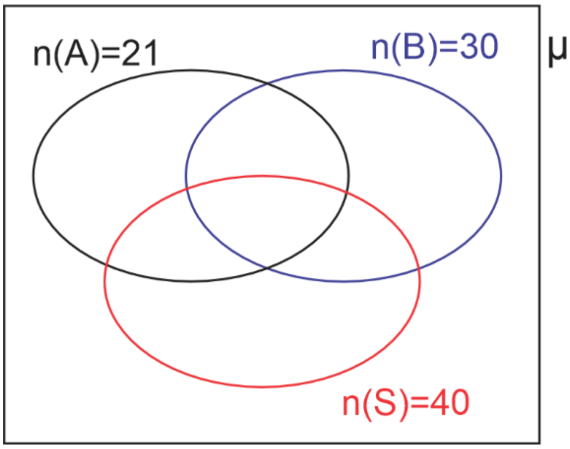
Find below the completed diagram:



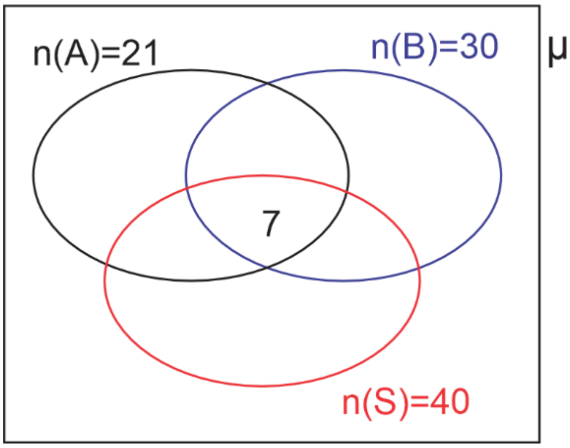

**Example 1.7**

A survey of 70 investors reveals the following: 30 invest in bonds, 40 in stocks and 21 in annuities. 10 had investments in bonds and annuities, 15 in bonds and stocks, 12 in stocks and annuities, and 7 had investments in all three assets. The remaining investors had investments in other assets.
Create a Venn diagram for this information.

**Solution**
First, draw a Venn diagram with three intersecting sets representing the three main sets: Bonds, Stocks, and Annuities. Let $B$ represent Bonds, $A$ represent Annuities and $S$ represent Stocks. This means $n(B) = 30$, $n(A) = 21$ and $n(S) = 40$.




Next, we fill in the region where all three sets intersect. From the question, this figure is 7.

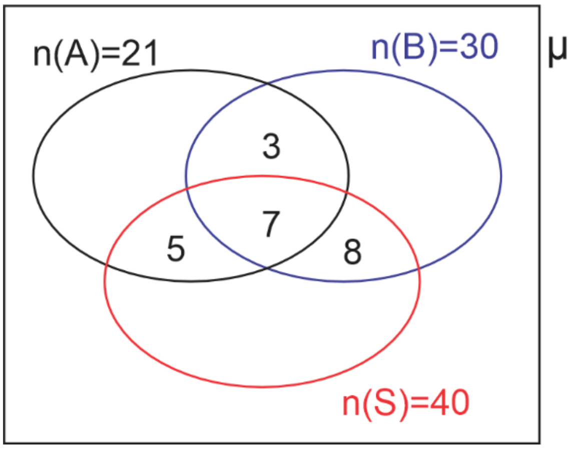



Let us add these values to the Venn diagram. Next, we fill in the regions where exactly two sets intersect:

a. **Bonds and Annuities.** A total of 10 had investments in bonds and annuities but 7 of them have investments in all three assets. This leaves $10 - 7 = 3$ investing in only Bonds and Annuities.

b. **Bonds and Stocks.** A total of 15 had investments in bonds and stocks but 7 of them had investments in all three assets. This leaves $15 - 7 = 8$ investing in only Bonds and Stocks.

c. **Stocks and Annuities.** A total of 12 had investments in stocks and annuities but 7 of them have investments in all three assets. This leaves $12 - 7 = 5$ investing in only Stocks and Annuities.



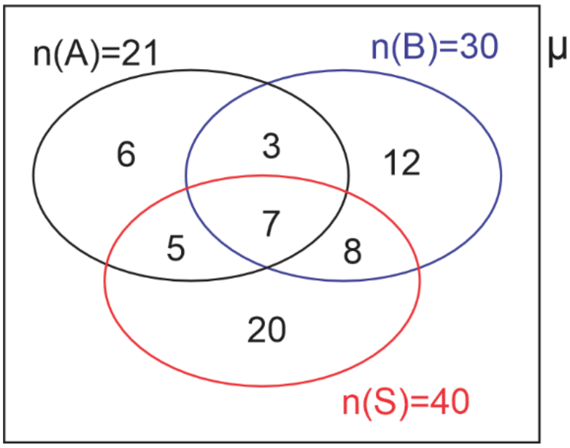

You will realise there is only one region remaining in each set. These are the regions that represent only one set:

a. **Only Bonds.** A total of 30 had investments in bonds but we have already accounted for $3 + 7 + 8 = 18$ of them, leaving $30 - 18 = 12$ having investments in only Bonds.

b. **Only Annuities.** A total of 21 had investments in annuities but we have already accounted for $3 + 7 + 5 = 15$ of them, leaving $21 - 15 = 6$ having investments in only Annuities.

c. **Only Stocks.** A total of 40 had investments in stocks but we have already accounted for $5 + 7 + 8 = 20$ of them, leaving $40 - 20 = 20$ investing in only Stocks.

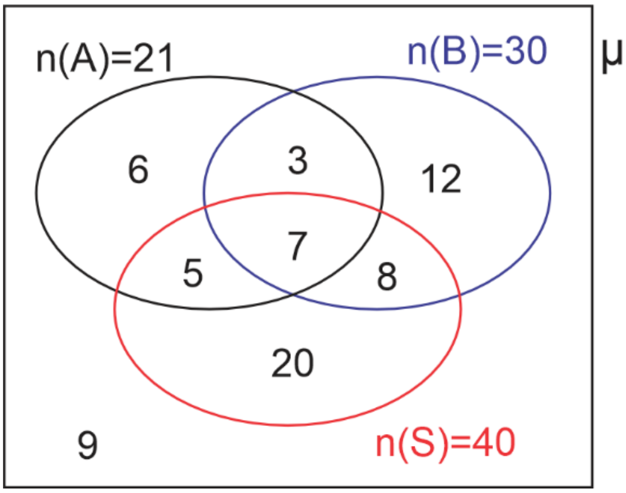
We will add these values to the Venn diagram:




Finally, we calculate the number of investors with no investments in the three assets. The sum of all the values already calculated for the intersecting sets is

$$12 + 3 + 6 + 8 + 5 + 7 + 20 = 61.$$

This means $70 - 61 = 9$ investors did not invest in any of the three assets. This value is recorded outside the circles but inside the universal set, completing the Venn diagram:




**Example 1.8**

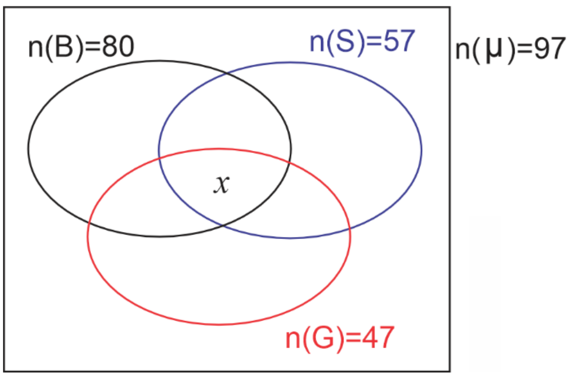
A survey of 97 countries that participated in the Olympics reveals that 80 won bronze, 57 won silver and 47 won gold. It also reveals that 30 won gold and silver, 50 won silver and bronze, 38 won gold and bronze, and 3 countries did not win any medal.

a. Illustrate the information on a Venn diagram.

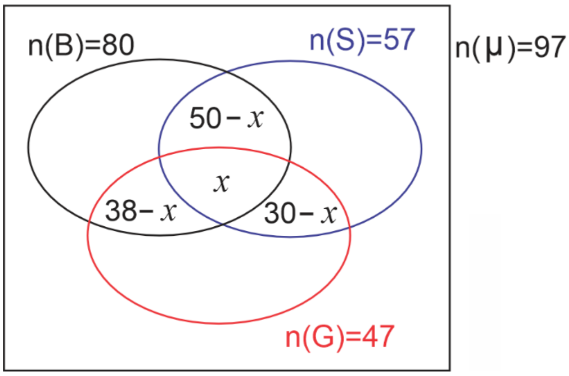
b. Calculate the number of countries who won all three medals.

c. Calculate the number of countries who won exactly two medals.

**Solution**
Let $B$ = Bronze medal, $G$ = Gold medal, and $S$ = Silver medal, so $n(B) = 80$, $n(S) = 57$ and $n(G) = 47$.
Let $x$ be the number of countries who won all three medals (unknown, since we were not given this directly).

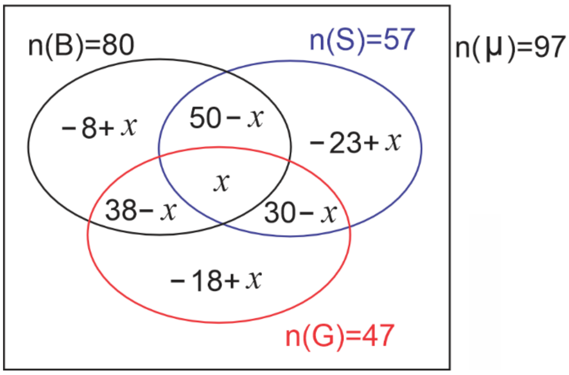



Next, we fill in the regions where exactly two sets intersect (countries that won exactly two of the three medals). A total of 30 countries won Gold and Silver; of these, $x$ won all three medals, so $(30-x)$ countries won *only* Gold and Silver. Similarly, 50 won Silver and Bronze, giving $(50-x)$ who won only Silver and Bronze, and 38 won Gold and Bronze, giving $(38-x)$ who won only Gold and Bronze.



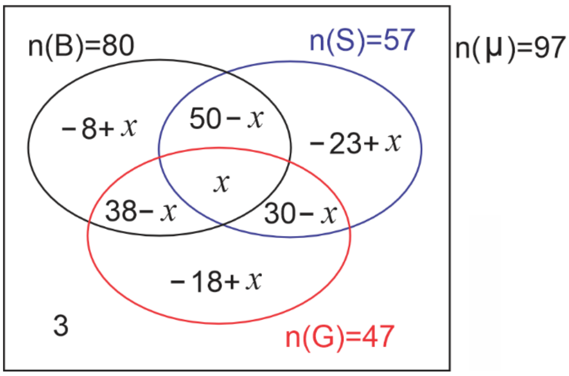

At this point only one region remains in each set:

- **Only Bronze.** 80 won bronze, and we have already accounted for $x + (50 - x) + (38 - x) = (88 - x)$ of them, leaving $80 - (88 - x) = (x - 8)$ winning only bronze.
- **Only Silver.** 57 won silver, and we have already accounted for $x + (50 - x) + (30 - x) = (80 - x)$ of them, leaving $57 - (80 - x) = (x - 23)$ winning only silver.
- **Only Gold.** 47 won gold, and we have already accounted for $x + (30 - x) + (38 - x) = (68 - x)$ of them, leaving $47 - (68 - x) = (x - 21)$ winning only gold.




*(Note: the Gold-only region in this diagram is labelled $-18 + x$ in the original textbook figure — this is a typo. As derived above, the correct formula is $x - 21$. We use the correct formula going forward.)*

According to the question, 3 countries did not win any medal, so $(G \cup S \cup B)' = 3$.




To solve for $x$, add all the entries and equate the result to the total number of countries under consideration (97):

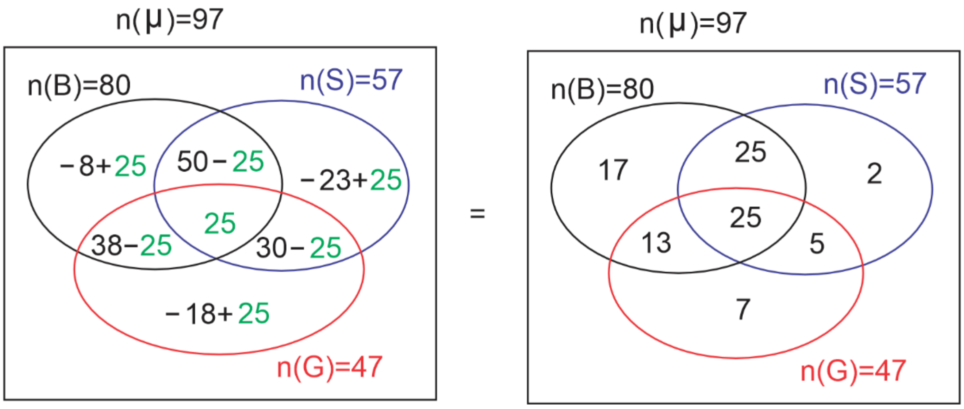
$$x + (30 - x) + (50 - x) + (38 - x) + (x - 8) + (x - 23) + (x - 21) + 3 = 97$$

$$x + 69 = 97 \implies x = 97 - 69 = 28.$$

e. Therefore, **28** countries won all three medals.

f. Number of countries that won exactly two medals $= (30-28)+(50-28)+(38-28) = 2 + 22 + 10 = 34$.

We can now update the diagram with this value. The completed Venn diagram, as given in the original textbook, is:




*Important note:* the textbook substitutes $x = 25$ into the region formulas above, not the value $x = 28$ that was just solved for algebraically. This is an error in the original — every region value shown (17, 25, 2, 13, 25, 5) other than the Gold-only region is inconsistent with $x = 28$, and the diagram's own values do not agree with part (f)'s answer of 34 for the number of countries winning exactly two medals ($25 + 13 + 5 = 43 \neq 34$, using the diagram's own printed numbers). Substituting the correctly-solved $x = 28$ into each formula instead gives Bronze-only $= x-8 = 20$, Bronze∩Silver-only $= 50-x = 22$, Silver-only $= x - 23 = 5$, Bronze∩Gold-only $= 38 - x = 10$, Silver∩Gold-only $= 30 - x = 2$, and Gold-only $= x - 21 = 7$ (triple intersection $= 28$), which does check against part (f): $2 + 22 + 10 = 34$.

## 3 Expanding Binomial Expressions

Hopefully you recall the work we did in Year 1 on the binomial expansion. We used the Pascal triangle and the Combination method to expand and simplify certain expressions. Let's remind you of this in the activity below.

**Activity 1.2: Revision**

1. In groups or pairs, write down or generate the Pascal triangle up to $(x + y)^5$.
2. Use your results in step 1 to simplify the expressions:
   a. $(2x + y)^3$
   b. $(a - 3b)^4$
   c. $(2 + \sqrt{x})^5$
3. Apply the combination method to also simplify the above tasks.
4. Compare your results in (2) and (3) and write down any observations.

5. Show your solutions, or write up, to classmates in other groups or to your mathematics teacher.

The Binomial Expansion helps us to expand and simplify complex expressions, which would have been difficult using the idea of algebraic expressions. Under algebraic expressions we could only expand with a positive integer power, but the binomial theorem will help us to expand any expression with a rational number as the power.

**Activity 1.3 – Derivation of Binomial Theorem**

1. Look for the ${}^nC_r$ button on your calculator.
2. Task: use ${}^nC_r$ on your calculator to evaluate
   i) ${}^3C_0$
   ii) ${}^3C_1$
   iii) ${}^3C_2$
   iv) ${}^3C_3$
3. Hopefully the answers you have are: 1 3 3 1. This was used in Year 1 to expand certain expressions.

The combination of $n$ items taking $r$ at a time is:

$$\binom{n}{r} = {}^nC_r = \frac{n!}{(n-r)!\,r!}, \qquad \text{where } n! \text{ is read as "}n\text{ factorial"}$$

and is explained as $n! = n(n-1)(n-2)(n-3)\cdots 2\times 1$.

For example, $5! = 5(5-1)(5-2)(5-3)(5-4) = 5\times 4\times 3\times 2\times 1 = 120$.

You could also find 5! using a calculator. Look for ! on your calculator to find 5!, to confirm the answer obtained above.

**Example 1.9**

In groups or in pairs, solve the following without using a calculator:

a. ${}^6C_3$

b. ${}^4C_2$

Confirm your answers with a calculator.

**Solution**

a. ${}^6C_3 = \dfrac{6!}{(6-3)!\,3!} = \dfrac{6\times 5\times 4\times 3\times 2\times 1}{3\times 2\times 1\times 3\times 2\times 1} = \dfrac{6\times 5\times 4}{3\times 2\times 1} = \dfrac{120}{6} = 20$

b. ${}^4C_2 = \dfrac{4!}{(4-2)!\,2!} = \dfrac{4\times 3\times 2\times 1}{2\times 1\times 2\times 1} = \dfrac{4\times 3}{2\times 1} = \dfrac{12}{2} = 6$

**Activity 1.4: General Form of Binomial Theorem**

1. Using the definition $\binom{n}{r} = {}^nC_r = \dfrac{n!}{(n-r)!\,r!}$, work in pairs or groups and discuss how the following evaluations were done:

a. $^nC_1 = \dfrac{n!}{(n-1)! \times 1!} = \dfrac{n(n-1)!}{(n-1)! \times 1} = n$

b. $^nC_2 = \dfrac{n!}{(n-2)! \times 2!} = \dfrac{n(n-1)(n-2)!}{(n-2)! \times 2 \times 1} = \dfrac{n(n-1)}{2}$

c. $^nC_3 = \dfrac{n!}{(n-3)! \times 3!} = \dfrac{n(n-1)(n-2)(n-3)!}{(n-3)! \times 3 \times 2 \times 1} = \dfrac{n(n-1)(n-2)}{6}$

2. Write down your challenges or observations for a class discussion with your classmates and teacher.

It must be noted that $^nC_0 = 1$ and $0! = 1$.

In general, the Binomial Theorem is written as:

$$(a+x)^n = {}^nC_0\, a^n x^0 + {}^nC_1\, a^{n-1}x^1 + {}^nC_2\, a^{n-2}x^2 + {}^nC_3\, a^{n-3}x^3 + \cdots + {}^nC_r\, a^{n-r}x^r + \cdots + {}^nC_n\, a^0 x^n$$

$$= a^n + n\,a^{n-1}x + \frac{n(n-1)}{2!}\,a^{n-2}x^2 + \frac{n(n-1)(n-2)}{3!}\,a^{n-3}x^3 + \cdots + x^n$$

There is a special case when $a = 1$. Substituting $a = 1$:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots + x^n$$

Let us go through the following worked examples to assist us in the expansion of the form $(1-x)^n$ or $(1+x)^n$.

**Example 1.10**

By using the Binomial Theorem, fully expand the expression $(1+x)^5$.

**Solution**

Using the binomial theorem,

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots + x^n$$

In this case $n = 5$, so we substitute this in:

$$(1+x)^5 = 1 + 5x + \frac{5 \times 4}{2!}x^2 + \frac{5 \times 4 \times 3}{3!}x^3 + \frac{5 \times 4 \times 3 \times 2}{4!}x^4 + x^5$$

Simplify the coefficients to obtain:

$$= 1 + 5x + \frac{5 \times 4}{2 \times 1}x^2 + \frac{5 \times 4 \times 3}{3 \times 2 \times 1}x^3 + \frac{5 \times 4 \times 3 \times 2}{4 \times 3 \times 2 \times 1}x^4 + x^5$$

$$= 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$$

**Example 1.11**

By using the Binomial Theorem, fully expand the expression $(1+2x)^4$.

**Solution**

Using the binomial theorem $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots + x^n$, in this case $n = 4$ and '$x$' $= 2x$, so we substitute these in:

$$(1+2x)^4 = 1 + 4(2x) + \frac{4 \times 3}{2!}(2x)^2 + \frac{4 \times 3 \times 2}{3!}(2x)^3 + (2x)^4$$

Simplify the coefficients to obtain:

$$= 1 + 4(2x) + \frac{4 \times 3}{2 \times 1} \times 4x^2 + \frac{4 \times 3 \times 2}{3 \times 2 \times 1} \times 8x^3 + 16x^4$$

$$= 1 + 8x + 24x^2 + 32x^3 + 16x^4$$

**Example 1.12**

Find the first four terms of the binomial $\sqrt{1+2x}$ using the binomial theorem.

**Solution**

$$\sqrt{1+2x} = (1+2x)^{1/2}$$

In this case $n = \frac{1}{2}$ and '$x$' $= 2x$, so we substitute these in:

$$\begin{aligned}(1+2x)^{1/2} &= 1 + \tfrac{1}{2}(2x) + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}(2x)^2 + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}(2x)^3 \\ &= 1 + \tfrac{1}{2}(2x) + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2 \times 1}(4x^2) + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3 \times 2 \times 1}(8x^3) \\ &= 1 + x - \tfrac{1}{2}x^2 + \tfrac{1}{2}x^3\end{aligned}$$

**Example 1.13**

Find the first four terms of the binomial $\sqrt[3]{1-3x}$ using the binomial theorem.

**Solution**

$$\sqrt[3]{1-3x} = (1-3x)^{1/3}$$

In this case $n = \frac{1}{3}$ and '$x$' $= -3x$, so we substitute these in:

$$\begin{aligned}(1-3x)^{1/3} &= 1 + \tfrac{1}{3}(-3x) + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}(-3x)^2 + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)\left(\frac{1}{3}-2\right)}{3!}(-3x)^3 \\ &= 1 + \tfrac{1}{3}(-3x) + \frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2 \times 1}(-3x)^2 + \frac{\frac{1}{3}\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3 \times 2 \times 1}(-3x)^3 \\ &= 1 + \tfrac{1}{3}(-3x) + \frac{-\frac{2}{9}}{2}(9x^2) + \frac{\frac{10}{27}}{6}(-27x^3) \\ &= 1 - x - x^2 - \tfrac{5}{3}x^3\end{aligned}$$

**Example 1.14**

Expand up to the fifth term the expression $\dfrac{1}{(1-x)^2}$.

**Solution**

Rewriting $\dfrac{1}{(1-x)^2}$ using the rules of indices, $\dfrac{1}{(1-x)^2} = (1-x)^{-2}$.

$$\begin{aligned}(1-x)^{-2} &= 1 - 2(-x) + \frac{-2(-2-1)}{2!}(-x)^2 + \frac{-2(-2-1)(-2-2)}{3!}(-x)^3 \\ &\quad + \frac{-2(-2-1)(-2-2)(-2-3)}{4!}(-x)^4 \\ &= 1 - 2(-x) + \frac{-2(-3)}{2\times 1}(-x)^2 + \frac{-2(-3)(-4)}{3\times 2\times 1}(-x)^3 + \frac{-2(-3)(-4)(-5)}{4\times 3\times 2\times 1}(-x)^4\end{aligned}$$

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + 5x^4$$

## Example 1.15

Find the first four terms of $(1-16x)^{1/4}$.

**Solution**

$$\begin{aligned}(1-16x)^{1/4} &= 1 + \tfrac{1}{4}(-16x) + \frac{\frac{1}{4}\left(\frac{1}{4}-1\right)}{2!}(-16x)^2 + \frac{\frac{1}{4}\left(\frac{1}{4}-1\right)\left(\frac{1}{4}-2\right)}{3!}(-16x)^3 \\ &= 1 + \tfrac{1}{4}(-16x) + \frac{\frac{1}{4}\left(-\frac{3}{4}\right)}{2\times 1}(-16x)^2 + \frac{\frac{1}{4}\left(-\frac{3}{4}\right)\left(-\frac{7}{4}\right)}{3\times 2\times 1}(-16x)^3 \\ &= 1 + \tfrac{1}{4}(-16x) + \frac{-\frac{3}{16}}{2}(256x^2) + \frac{\frac{21}{64}}{6}(-4096x^3) \\ &= 1 - 4x - 24x^2 - 224x^3\end{aligned}$$

## Activity 1.5: Deciding Whether the Binomial Theorem Can Be Used in All Cases

Discuss in groups whether the binomial theorem

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots + x^n$$

can be used in all cases.

1. Consider $(2+4x)^4$. Can the binomial theorem be used directly?
2. How will you write the terms in the bracket to have 1 as one of the terms?
3. Let us go through the following tasks to help us write $(2+4x)^4$ in the form $(1+x)^n$ and use the binomial theorem to expand and simplify:

   **Task 1:** Take the given binomial expression $(2+4x)$.

   **Task 2:** Factorise the constant out from the binomial to obtain $2(1+2x)$.

   **Task 3:** Rewrite your expression obtained with the power as given, i.e. $2^4(1+2x)^4$.

   **Task 4:** Expand the binomial $(1+2x)^4$, after which you multiply your result by $2^4$.

   Now follow the tasks to obtain all the terms for the given expression.

   **Expected Answer:** $16 + 128x + 348x^2 + 512x^3 + 256x^4$.

Let's solve the following examples using the binomial theorem.

**Example 1.16**

1. Use the binomial theorem to expand fully $(3+9x)^5$.
2. Find the first three terms of $\left(\frac{4}{9}+6x\right)^{1/2}$ using the binomial theorem.
3. Use the binomial theorem to expand fully and simplify $(10+4x)^3$.
4. Use the binomial theorem to find the first four terms of the expression $\sqrt[4]{81+162x}$.

**Solution**

**1.**

$$(3+9x)^5 = 3^5(1+3x)^5$$

$$\begin{aligned}3^5(1+3x)^5 = 3^5\Bigg[&1+5(3x)+\frac{5(4)}{2\times 1}(3x)^2+\frac{5(4)(3)}{3\times 2\times 1}(3x)^3\\ &+\frac{5(4)(3)(2)}{4\times 3\times 2\times 1}(3x)^4+\frac{5(4)(3)(2)(1)}{5\times 4\times 3\times 2\times 1}(3x)^5\Bigg]\end{aligned}$$

$$\begin{aligned}&= 243\left[1+5(3x)+10(9x^2)+10(27x^3)+5(81x^4)+(243x^5)\right]\\ (3+9x)^5 &= 243\left(1+15x+90x^2+270x^3+405x^4+243x^5\right)\end{aligned}$$

Compute the final coefficients:

- $243\times 1 = 243$
- $243\times 15 = 3645$
- $243\times 90 = 21870$
- $243\times 270 = 65610$
- $243\times 405 = 98415$
- $243\times 243 = 59049$

Therefore, the expansion is

$$(3+9x)^5 = 243+3645x+21870x^2+65610x^3+98415x^4+59049x^5.$$

**2.**

$$\begin{aligned}\left(\frac{4}{9}+6x\right)^{1/2} &= \left(\frac{4}{9}\right)^{1/2}\left(1+\frac{27}{2}x\right)^{1/2}\\ &= \left(\frac{4}{9}\right)^{1/2}\left[1+\frac{1}{2}\left(\frac{27}{2}x\right)+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(\frac{27}{2}x\right)^2\right]\\ &= \frac{2}{3}\left[1+\frac{27}{4}x+\frac{-\frac{1}{4}}{2}\left(\frac{729}{4}x^2\right)\right] = \frac{2}{3}\left[1+\frac{27}{4}x-\frac{729}{32}x^2\right]\\ &= \frac{2}{3}+\frac{9}{2}x-\frac{243}{16}x^2\end{aligned}$$

**3.**

$$(10+4x)^3 = 10^3\left(1+\frac{2}{5}x\right)^3$$

$$= 10^3\left[1+3\left(\frac{2}{5}x\right)+\frac{3(2)}{2\times 1}\left(\frac{2}{5}x\right)^2+\frac{3(2)(1)}{3\times 2\times 1}\left(\frac{2}{5}x\right)^3\right]$$

$$= 10^3\left[1+\frac{6}{5}x+3\left(\frac{4x^2}{25}\right)+\frac{8}{125}x^3\right] = 1000\left[1+\frac{6}{5}x+\frac{12x^2}{25}+\frac{8x^3}{125}\right]$$

$$(10+4x)^3 = 1000+1200x+480x^2+64x^3$$

**4.**

$$\sqrt[4]{81+162x} = (81+162x)^{1/4} = 81^{1/4}(1+2x)^{1/4}$$

$$= 81^{1/4}\left[1+\frac{1}{4}(2x)+\frac{\frac{1}{4}\left(\frac{1}{4}-1\right)}{2!}(2x)^2+\frac{\frac{1}{4}\left(\frac{1}{4}-1\right)\left(\frac{1}{4}-2\right)}{3!}(2x)^3\right]$$

$$= 81^{1/4}\left[1+\frac{1}{4}(2x)+\frac{-\frac{3}{16}}{2\times 1}(4x^2)+\frac{\frac{21}{64}}{3\times 2\times 1}(8x^3)\right] = 3\left[1+\frac{1}{2}x-\frac{3}{8}x^2+\frac{7}{16}x^3\right]$$

$$\sqrt[4]{81+162x} = 3+\frac{3}{2}x-\frac{9}{8}x^2-\frac{21}{16}x^3$$

**Example 1.17**

The first three terms in the expansion of $\left(1+\frac{x}{p}\right)^n$ in ascending powers of $x$ are $1+x+\frac{9}{20}x^2$. Find the values of $n$ and $p$.

**Solution**

Expand $\left(1+\frac{x}{p}\right)^n$ using the binomial theorem:

$$\left(1+\frac{x}{p}\right)^n = 1+n\frac{x}{p}+\frac{n(n-1)}{2!}\left(\frac{x}{p}\right)^2 = 1+nx+\frac{n(n-1)}{2}\cdot\frac{x^2}{p^2}$$

Equating this to $1+x+\frac{9}{20}x^2$ and comparing the coefficients of $x$ and $x^2$:

$$\frac{n}{p}=1, \qquad \frac{n(n-1)}{2p^2}=\frac{9}{20}$$

$$n=p \quad \ldots(1) \qquad\qquad 20n(n-1)=18p^2 \quad \ldots(2)$$

Substituting $n=p$ from (1) into (2):

$$20p(p-1)=18p^2 \implies 20p^2-20p=18p^2 \implies 2p^2=20p \implies p=10$$

$$\therefore\ p=10, \qquad n=10$$

**Example 1.18**

Find the values of $t$ and $n$ in the equation $(1+tx)^n = 1 - 6x + \frac{33}{2}x^2$.

**Solution**
Expand $(1+tx)^n$ using the binomial theorem:

$$(1+tx)^n = 1 + ntx + \frac{n(n-1)}{2!}(tx)^2 = 1 + ntx + \frac{n(n-1)}{2}t^2x^2$$

Equating this to $1 - 6x + \frac{33}{2}x^2$ and comparing coefficients of $x$ and $x^2$:

$$nt = -6 \quad \ldots(1) \qquad\qquad \frac{n(n-1)}{2}t^2 = \frac{33}{2} \implies 2n(n-1)t^2 = 66 \quad \ldots(2)$$

Rewriting (2): $(2n^2 - 2n)t^2 = 66 \implies 2(nt)^2 - 2(nt)t = 66$. Since $nt = -6$:

$$2(-6)^2 - 2(-6)t = 66 \implies 72 + 12t = 66 \implies 12t = -6 \implies t = -\frac{1}{2}$$

Substituting back into $nt = -6$:

$$n\left(-\frac{1}{2}\right) = -6 \implies -n = -12 \implies n = 12$$

**Example 1.19**

Expand $(1+2x-x^2)^6$ as far as the $x^2$ term.

**Solution**
How will you use the binomial theorem to solve a question of this nature? Discuss your views with a classmate or in your groups.
You need to write the trinomial $1 + 2x - x^2$ as a binomial in the form $(1+x)^n$. We represent $2x - x^2$ by a variable, say $y$, i.e. $y = 2x - x^2$, so

$$(1+2x-x^2)^6 = (1+y)^6.$$

Expanding $(1+y)^6$ using the binomial theorem:

$$(1+y)^6 = 1 + 6y + \frac{6(6-1)}{2!}y^2 + \cdots = 1 + 6y + \frac{6(5)}{2}y^2 = 1 + 6y + 15y^2$$

But $y = 2x - x^2$, so substituting:

$$= 1 + 6(2x - x^2) + 15(2x - x^2)^2 + \cdots$$

$$\begin{aligned} &= 1 + 6(2x - x^2) + 15(4x^2 - 4x^3 + x^4) + \cdots \\ &= 1 + 12x - 6x^2 + 60x^2 - 60x^3 + 15x^4 + \cdots \\ (1+2x-x^2)^6 &= 1 + 12x + 54x^2 + \cdots \end{aligned}$$

**Example 1.20**

Expand $(1-3x+x^2)^5$ as far as the $x^3$ term.

**Solution**
We represent $-3x+x^2$ by a variable, say $y$, i.e. $y=-3x+x^2$, so

$$(1-3x+x^2)^5=(1+y)^5.$$

Expanding $(1+y)^5$ using the binomial theorem:

$$(1+y)^5=1+5y+\frac{5(4)}{2}y^2+\frac{5(4)(3)}{6}y^3=1+5y+10y^2+10y^3$$

But $y=-3x+x^2$, so substituting:

$$=1+5(-3x+x^2)+10(-3x+x^2)^2+10(-3x+x^2)^3$$

$$=1+5(-3x+x^2)+10(9x^2-6x^3+x^4)+10(-27x^3+27x^4-9x^5+x^6)$$
$$=1-15x+5x^2+90x^2-60x^3+10x^4-270x^3+270x^4-90x^5+10x^6$$

Collecting terms up to $x^3$:

$$(1-3x+x^2)^5=1-15x+95x^2-330x^3+\cdots$$

Did you know we can use binomial expansion to approximate exponential numbers? This powerful technique allows us to simplify complex expressions and make calculations easier. We will explore how to apply binomial expansion to approximate exponential functions.

# 4 Applying Binomial Expansion to Approximate Exponential Numbers

Did you know we can use binomial expansion to approximate exponential numbers? This powerful technique allows us to simplify complex expressions and make calculations easier. We will explore how to apply binomial expansion to approximate exponential functions.

**Activity 1.6: Approximating Exponential Numbers Using Binomial Expansion $(a+b)^n$**

Ensure you have your notebook, calculator and pen ready for the activity.

1. Choose a number to work with. Examples include: 1.5, 3.5, 10, etc.
2. Express your chosen number in the form $(a+b)$. For example: if you chose 1.5, you could write it as $(1+0.5)$.
3. Select any integer to represent $n$. Examples include: 2, 4, $-4$.
4. Write your decimal and $n$ in the form $(a+b)^n$. For example, using 1.5 and $n=4$, you would write $(1+0.5)^4$.
5. Apply the binomial theorem to expand and evaluate your expression.
6. Choose different sets of numbers and repeat the steps above. Experiment with various decimal values and $n$ to see how the approximations change.

7. Use your calculator to find the actual value of your chosen decimal raised to the power of $n$. For example, calculate $1.5^4$.
8. Look at the results from your binomial expansion and the calculator.
9. Discuss:
   a. How close were your approximations to the actual values?
   b. What did you observe about the accuracy of your approximations with different $n$ values?

Great work! You have effectively used binomial expansion to approximate exponential numbers. Now, let us solve more examples together to deepen your understanding!

**Example 1.21**

Expand $(1+2y)^5$ using the binomial theorem. Hence use your results to evaluate $(1.04)^5$, correct to 4 decimal places.

**Solution**

$$\begin{aligned}(1+2y)^5 &= 1+5(2y)+\frac{5(4)}{2\times 1}(2y)^2+\frac{5(4)(3)}{3\times 2\times 1}(2y)^3\\ &\quad+\frac{5(4)(3)(2)}{4\times 3\times 2\times 1}(2y)^4+\frac{5(4)(3)(2)(1)}{5\times 4\times 3\times 2\times 1}(2y)^5\\ &= 1+5(2y)+10(4y^2)+10(8y^3)+5(16y^4)+1(32y^5)\\ (1+2y)^5 &= 1+10y+40y^2+80y^3+80y^4+32y^5\end{aligned}$$

$(1.04)^5$ can be written as $(1+0.04)^5$. Comparing $(1+0.04)^5$ to $(1+2y)^5$:

$$2y = 0.04 \implies y = 0.02$$

Substituting $y = 0.02$ into the expansion:

$$1+10(0.02)+40(0.02)^2+80(0.02)^3+80(0.02)^4+32(0.02)^5$$

$$= 1+0.2+0.016+0.00064+0.0000128+0.0000001024 = 1.2166529024$$

$$\approx 1.2167 \text{ (to 4 d.p.)}$$

Thus $(1.04)^5 \approx 1.2167$.

**Example 1.22**

Find the first five terms of the expression $\sqrt{1+3x}$. Hence find an estimate of the value of $\sqrt{1.12}$ to 5 decimal places.

**Solution**

$$\sqrt{1+3x} = (1+3x)^{1/2}$$

$$
\begin{aligned}
(1+3x)^{1/2} &= 1+\frac{1}{2}(3x)+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}(3x)^2+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}(3x)^3 \\
&\quad+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\left(\frac{1}{2}-3\right)}{4!}(3x)^4 \\
&= 1+\frac{3x}{2}+\frac{-\frac{1}{4}}{2}(3x)^2+\frac{\frac{3}{8}}{6}(3x)^3+\frac{-\frac{15}{16}}{24}(3x)^4 \\
&= 1+\frac{3x}{2}-\frac{1}{8}(9x^2)+\frac{1}{48}(27x^3)-\frac{5}{128}(81x^4)
\end{aligned}
$$

$$\sqrt{1+3x}=1+\frac{3x}{2}-\frac{9x^2}{8}+\frac{27x^3}{16}-\frac{405x^4}{128}$$

$\sqrt{1.12}$ can be written as $(1+0.12)^{1/2}$. Comparing $(1+0.12)^{1/2}$ to $(1+3x)^{1/2}$:

$$3x=0.12 \implies x=0.04$$

Substituting $x=0.04$ into the expansion:

$$
\begin{aligned}
&1+\frac{3(0.04)}{2}-\frac{9(0.04)^2}{8}+\frac{27(0.04)^3}{16}-\frac{405(0.04)^4}{128} \\
&= 1+\frac{0.12}{2}-\frac{0.0144}{8}+\frac{0.001728}{16}-\frac{0.00020736}{128} \\
&= 1+0.06-0.0018+0.000108-0.0000081=1.0582999 \\
&\approx 1.05830 \text{ (to 5 d.p.)}
\end{aligned}
$$

Thus $\sqrt{1.12}\approx 1.05830$.

**Example 1.23**

Using the binomial theorem, find an estimate for the value of $\sqrt{101}$.

**Solution**

$$\sqrt{101}=\sqrt{100+1}=(100+1)^{1/2}=100^{1/2}\left(1+\frac{1}{100}\right)^{1/2}=10\,(1+0.01)^{1/2}$$

$$
\begin{aligned}
10(1+0.01)^{1/2} &= 10\left[1+\frac{1}{2}(0.01)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(0.01)^2+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(0.01)^3+\cdots\right] \\
&= 10\left[1+\frac{1}{2}(0.01)-\frac{1}{8}(0.0001)+\frac{1}{16}(0.000001)\right] \\
&= 10\,[1+0.005-0.0000125+0.0000000625] \\
&= 10\,[1.004987563]=10.04987563
\end{aligned}
$$

Thus $\sqrt{101}\approx 10.0499$.

**Example 1.24**

Use the binomial theorem to find approximations for:

a. $(3.97)^{1/2}$

b. $(0.994)^{1/4}$

c. $\sqrt{25.05}$

**Solution**

**a.**

$$(3.97)^{1/2} = (4 - 0.03)^{1/2} = 4^{1/2}(1 - 0.0075)^{1/2} = 2(1 - 0.0075)^{1/2}$$

$$\begin{aligned} 2(1 - 0.0075)^{1/2} &= 2\left[1 + \frac{1}{2}(-0.0075) + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(-0.0075)^2 + \cdots\right] \\ &= 2\left[1 - 0.00375 - \frac{1}{8}(0.0075)^2 + \cdots\right] \\ &= 2\left[1 - 0.00375 - 0.00003125 + \cdots\right] \\ &= 2\left[0.99621875\right] = 1.9924375 \end{aligned}$$

$$\therefore (3.97)^{1/2} \approx 1.992 \text{ (to 4 s.f.)}$$

**b.**

$$(0.994)^{1/4} = (1 - 0.006)^{1/4}$$

$$\begin{aligned} (1 - 0.006)^{1/4} &= 1 + \frac{1}{4}(-0.006) + \frac{\frac{1}{4}\left(-\frac{3}{4}\right)}{2!}(-0.006)^2 + \cdots \\ &= 1 - 0.0015 - \frac{1}{32}(0.000036) + \cdots \\ &= 1 - 0.0015 - 0.000001125 + \cdots \\ &= 0.998498875 \end{aligned}$$

$$\therefore (0.994)^{1/4} \approx 0.9985 \text{ (to 4 s.f.)}$$

**c.**

$$\sqrt{25.05} = (25 + 0.05)^{1/2} = 25^{1/2}(1 + 0.002)^{1/2} = 5(1 + 0.002)^{1/2}$$

$$\begin{aligned} 5(1 + 0.002)^{1/2} &= 5\left[1 + \frac{1}{2}(0.002) + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(0.002)^2 + \cdots\right] \\ &= 5\left[1 + 0.001 - \frac{1}{8}(0.000004) + \cdots\right] \\ &= 5\left[1 + 0.001 - 0.0000005\right] = 5\left[1.0009995\right] \\ &= 5.0049975 \end{aligned}$$

$$\therefore \sqrt{25.05} \approx 5.005 \text{ (to 4 s.f.)}$$

## Extended Reading

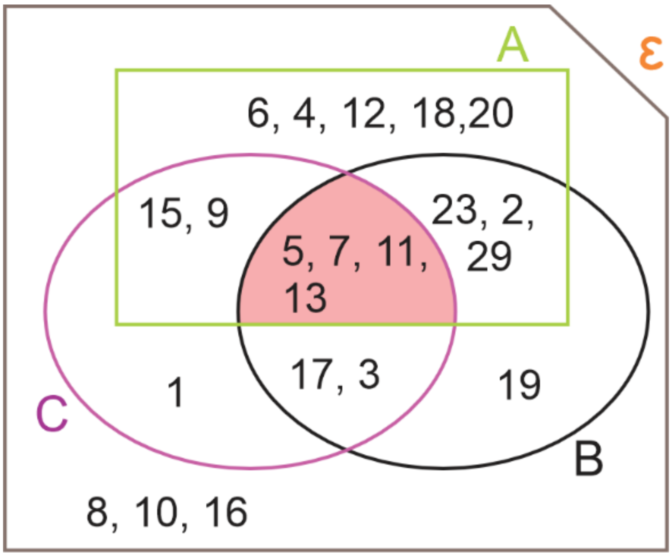

- Cambridge Additional Mathematics by Michael Haese, Sandra Haese, Mark Humphries, Chris Sangwin. Haese Mathematics (2014).
- Goldie, S. (2012). *Pure Mathematics.* Hodder Education.
- Talbert, J. F., & Heng, H. H. (1995). *Additional Mathematics: Pure & Applied.* Pearson

Education South Asia.

## Review Questions

### Part A

1. Use the Venn diagram below to answer the questions.




a. List the elements in $A$, $B$ and $C$.

b. List the elements in $A'$, $B'$ and $C'$.

c. Find (i) $(A \cup B \cup C)'$ (ii) $A' \cap B' \cap C'$ (iii) $(A \cap B \cap C)'$ (iv) $A' \cup B' \cup C'$.

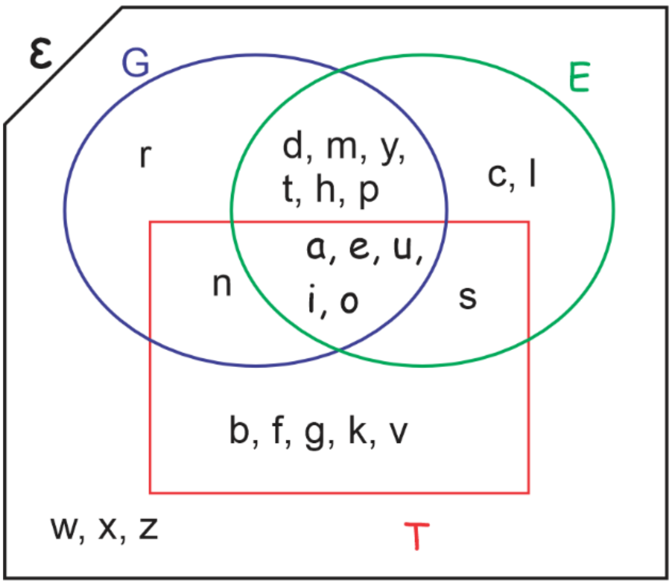
d. Compare your result in (c)(i) to (c)(ii). What conclusion can you draw?

e. Compare your result in (c)(iii) to (c)(iv). What conclusion can you draw?

f. Describe set $B$.

g. Describe set $C$.

2. Study the diagram carefully and use it to answer the questions.



a. List the elements in $G$, $E$, $T$ and $\varepsilon$.
b. Show that $G' \cap E' = (G \cup E)'$.
c. Show that $(G \cap E \cap T)' = G' \cup E' \cup T'$.
d. Show that $(G \cup E \cup T)' = G' \cap E' \cap T'$.

For questions 3–7, sets $X$, $Y$ and $Z$ are subsets of the universal set $\mu$. Use the information to choose the correct answer to the questions.

3. Find $X \cup X'$.
   **A.** $\varnothing$ **B.** $\mu$ **C.** $X$ **D.** $Y$

4. Find $Z \cap Z'$.
   **A.** $\varnothing$ **B.** $\mu$ **C.** $Z$ **D.** $X$

5. Find $(Y')'$.
   **A.** $\varnothing$ **B.** $\mu$ **C.** $X$ **D.** $Y$

6. Find $(X \cup X')'$.
   **A.** $\varnothing$ **B.** $\mu$ **C.** $X$ **D.** $X' \cup X$

7. Find $(Z \cap Z')'$.
   **A.** $\varnothing$ **B.** $\mu$ **C.** $X \cup Y$ **D.** $Z' \cap Z$

8. Rewrite:
   a. $(R \cup S')'$
   b. $(M' \cap N)'$
   c. $[(A' \cup B)' \cap C]'$
   d. $[(P \cap Q')' \cup P]'$
   e. $[A \cap (A' \cup B)']'$

9. Show that
   a. $[(R \cup M)' \cap P]' = R \cup M \cup P'$
   b. $[(R \cap M) \cup M']' = R' \cap M$

10. On a Venn diagram, shade the region represented by:
    a. $(A \cap B)' \cap C$
    b. $A' \cup (Y \cap Z')$

11. Sets $X = \{\text{prime numbers}\}$, $Y = \{W : 0 < W \leq 11\}$ and $Z = \{1, 2, 3, 4, 5, 6, 7, 13, 14, 15, 16, 18\}$ are subsets of $A = \{0, 1, 2, 3, 4, \ldots, 21\}$.
    a. List the members of the following sets:
       i. $X$
       ii. $Y$
       iii. $X \cap Y \cap Z$
       iv. $X \cup Y'$
       v. $(X \cup Y) \cap Z'$
       vi. $(X' \cap Y) \cup Z$
    b. Use a Venn diagram to represent $X$, $Y$, $Z$ and $A$.

12. A survey of 90 teachers reveals the following. 45 teach Management, 57 Accounting and 50 Costing. 25 teach Management and Accounting, 35 teach Accounting and Costing and 22 teach Management and Costing. 5 teachers cannot teach any of the three subjects.

    a. Represent the information on a Venn diagram.

    b. Calculate the number of teachers who can teach only one of the three subjects.

13. A survey of 41 employees shows that 25 are skilled in Carpentry, 15 in Brick laying and 24 in Tiling. 12 are skilled in carpentry and brick laying, 9 in brick laying and tiling, 17 in carpentry and tiling, and 8 are not skilled in any of the three occupations. Find the number of employees who are skilled in

    a. all three occupations

    b. exactly two of the occupations

14. A class of 60 students were each required to have textbooks in Accounting, Management and Costing. However, when the class teacher checked their books, she found out that: 16 had Accounting and Management books, 14 had Accounting and Costing books, 18 had only two of the three books and 20 had Accounting plus at least one other book. Every student had at least one of the three books. Find the number of students who had

    a. all three books.

    b. only Costing and Accounting books.

    c. exactly one of the three books.

    d. If 16 students had neither Accounting nor Costing books, find the probability that a student selected at random from the class had Management books.

15. The analysis of results of a sample of students who sat for the WASSCE in Biology, Physics and Mathematics revealed that: 80% passed in Biology, 65% passed in Mathematics, 70% passed in Physics, 52% passed in Biology and Mathematics, 58% passed in Biology and Physics, 55% passed in Physics and Mathematics, 45% passed in all three subjects. What percentage of the students:

    a. passed only Mathematics

    b. passed only Physics

    c. passed only Biology

    d. passed Biology and Physics but not Mathematics

    e. did not pass any of the subjects.

16. A shop owner kept records of purchases made by customers over the weekend. The shop sells food, clothing and hardware. She found that, out of the customers who visited her shop, 210 purchased food products, 100 hardware and 147 clothing. 30 purchased food and hardware. 128 purchased two of the three products. She also found that of those who purchased two of the three products, 14 did not buy clothing and 20 did not buy food. 13 of the customers who visited the store did not buy any of the products.

    a. How many customers visited the store over the weekend?

    b. How many of the customers who visited the store:

        i. purchased less than two products.

        ii. purchased all three products.

        iii. purchased food and other products.

17. A group of 210 tourists were asked to indicate which of the languages, German, French and English they can speak. It was found that 55 can speak German, 76 can speak French and 200 can speak English. 103 can speak two of the three languages and 6 tourists cannot speak any of the three languages. Find the number of tourists who can speak:

    a. all three languages

    b. only one of the three languages.

18. 87 schools in Tamale have subscriptions to at least one of the following newspapers: Graphic, Mirror and Times. 69 subscribe to Graphic, 62 to Mirror and 55 to Times. 32 have subscriptions to all three newspapers. Find the number of schools who have subscriptions to:

    a. exactly two of the newspapers

    b. only one of the newspapers.

19. A publishing company surveyed its employees about their proficiency in three software: Photoshop, CorelDraw and InDesign. The results show that all the employees were proficient in at least one of the three software and 10 were proficient in all three. The number of employees proficient in CorelDraw and other software were twice those proficient in only CorelDraw. Of the employees who were proficient in two of the software, 14 were not proficient in InDesign, 8 were not proficient in CorelDraw and 6 were not proficient in Photoshop. The same number of employees proficient in only InDesign are proficient in only Photoshop. There were more employees proficient in only Photoshop and InDesign than employees proficient in only Photoshop and there were fewer employees proficient in only InDesign and CorelDraw than employees proficient in only InDesign.

    a. Use a Venn diagram to represent the above data.

    b. How many employees were involved in the survey?

    c. How many employees were proficient in only one of the software?

## Part B

1. Write down the first five terms of the binomial expansion of:

    a. $(1+4x)^{12}$

    b. $(1-3x)^{9}$

    c. $(3+2x)^{7}$

    d. $\left(1-\dfrac{x}{2}\right)^{8}$

    e. $\left(2-\dfrac{5}{2}x\right)^{6}$

2. Write down the term indicated in the binomial expansion of the following functions:

    a. $(1+3x)^{6}$: 3rd term

    b. $\left(1+\dfrac{x}{3}\right)^{20}$: 7th term

    c. $(a+4b)^{12}$: term containing $a^4$

3. Expand the following binomial expansions up to and including the terms in $x^2$:

    a. $(1-2x)(1+x)^{8}$

b. $(2+3x)(1+3x)^{10}$

c. $(5+2x)\left(1-\frac{2}{3}x\right)^{12}$

d. $(1+4x)(1-3x)^{20}$

4. Expand $(1-2x)^7$ and hence find the value of $(0.98)^7$ correct to 4 decimal places.

5. Find the first four terms of the following using the binomial theorem.

a. $\left(1-\frac{1}{2}x\right)^{1/2}$

b. $(1-2x)^{-3}$

c. $(3-x)^{-2}$

6. Use appropriate binomial expansions to find the approximate values of:

a. $\sqrt[3]{8.7}$ correct to 5 significant figures.

b. $(81.162)^{1/4}$ to 4 decimal places.

c. $\sqrt{4.02}$ to 3 decimal places.

d. $\frac{1}{\sqrt{10}}$ to 4 decimal places.

e. $\sqrt{101}$ to 4 decimal places.

f. $(0.995)^5$ to 4 decimal places.

7. Show that in ascending powers of $x$, $\left(\frac{1-x}{1+x}\right)^{1/3}$ is $1-\frac{2}{3}x+\frac{2}{9}x^2-\frac{4}{21}x^3-\cdots$.

8. Expand the following as far as the $x^3$ term:

a. $(1+x+2x^2)^3$

b. $(1+3x+x^2)^4$

9. a. Find the first 3 terms in ascending powers of $x$ of the binomial expansion $(1+py)^9$, where $p$ is a non-zero constant.

b. If the first 3 terms are 1, $36x$ and $qx^2$ where $q$ is a constant, find the value of $p$ and $q$.

10. a. Find the first 3 terms in ascending powers of $x$ of the binomial expansion $(2+kx)^7$, where $k$ is a constant.

b. Given that the coefficient of $x^2$ is 6 times the coefficient of $x$, find the value of $k$.